

Formulas ,Functions and charts in Excel

FORMULAS AND FUNCTIONS

To manipulate data and to extract useful information from Excel worksheets, formulas and worksheet functions play very important role. In Excel, formulas are used to calculate results from the worksheet data. When there is some change in the data, such formulas automatically calculate the updated results with no extra efforts on the part of the user. There is a new feature introduced by Excel 2007, which enables you to create formulas which use columns names from a table, when you are working with table. This feature helps the user to make formulas much easier to read.

A formula can have any or all of the following elements

- Must begin with the 'equal to' = sign.
- Mathematical operators, such as + (for addition) and / (for division) and logical operators such as <, >
- References of cell (including named ranges and cells)
- Text or Values
- Functions related to the worksheets, for example SUM or AVERAGE

The current cell in which you have entered a formula will display the result after the formula is completely entered. Also, when you select or click on a cell which is having some formula, the formula will appear in the formula bar.

In Excel 2007, the formulas are available in the Formulas Tab. If you click on the Formulas tab, you can see the corresponding ribbon display with available formulas.

Copying a formula In above example just drag the handle and bring down to cover the remaining cells in the column total. This will automatically copy the formula and calculate the corresponding sum of the respective rows of the other students.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Name	Physics	Chemistry	Maths	English	Total															
2	Asmita	99	76	89	78	342															
3	Ashok	88	78	78	76	320															
4	Rahul	77	91	55	80	303															
5	Anurag	88	54	80	76	298															
6	Gaurav	77	79	89	87	332															
7	Prashant	45	59	78	87	269															
8	Tanavi	89	80	67	76	312															
9	Priyanka	47	55	89	67	258															
10	Monika	65	70	87	87	309															
11	Neha	94	89	76	88	347															
12																					
13																					
14																					
15																					
16																					
17																					
18																					
19																					
20																					
21																					
22																					

Some other examples of formulaes:

If A1 = 2, B1 = 3, C1 = 4, D1 = 5 then

(i) = B1 * C1 will give result as 3 * 4 = 12

(ii) = A1 * B1 – C1 + D1 will give result as 2 * 3 – 4 + 5 = 7

(iii) = ((A1 * B1) + C1)/D1 will give result as ((2 * 3) + 4)/5 = (6 + 4)/5 = 10/5 = 2

Auto sum feature

The sum of cell values can also be done with the help of AutoSum feature also.

Following steps explains how to do that.

Click on the cell F3.

Select the Formulas tab

Click AutoSum from the function library group

Select Sum

Press Enter

Alternatively, you can write = SUM(B3:E3) in the Cell F3 to get the sum of the Physics, Chemistry, Maths and English marks.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Name	Physics	Chemistry	Maths	English	Total															
2	Asmita	99	76	89	78	342															
3	Ashok	88	78	78	76	320															
4	Rahul	77	91	55	80	303															
5	Anurag	88	54	80	76	298															
6	Gaurav	77	79	89	87	332															
7	Prashant	45	59	78	87	269															
8	Tanavi	89	80	67	76	312															
9	Priyanka	47	55	89	67	258															
10	Monika	65	70	87	87	309															
11	Neha	94	89	76	88	347															
12																					
13																					
14																					
15																					
16																					
17																					
18																					
19																					
20																					
21																					
22																					

Apart from sum there are other functions also, like average, Count Numbers, Max, Min and many other Functions.

SUMIF(range, criteria, sum_range) : This form of sum functions is used to add the cells with respective to a given criteria.

Range : A group of adjacent cells

Cells within a range must be numbers or names, arrays, or references which are having numbers. In sumif function the blank and text values are ignored.

Criteria: It can be a number, expression, or text to define which cells will be considered for the addition e.g. criteria can be fixed as 12, "12", ">12", or "mangoes".

Sum_range: These are the actual cells to be added if their corresponding cells in range match criteria. In case the sum_range is avoided, then the cells in range are both evaluated by criteria and added if they match criteria.

SUMIF function, step by step:

1. Select the cell F3
2. Type =SUMIF
3. Double click the SUMIF command

	A	B	C	D	E	F	G	H	I	J	K
1	Name	Type 1	Total								
2	Bulbasaur	Grass	318		Type	Total Sum					
3	Ivysaur	Grass	405		Grass	=SUMIF					
4	Venusaur	Grass	525		Fire	SUMIF					
5	Charmander	Fire	309		Water	SUMIFS					
6	Charmeleon	Fire	405								
7	Charizard	Fire	534								
8	Squirtle	Water	314								
9	Wartortle	Water	405								
10	Blastoise	Water	530								
11											

- Specify the range for the condition B2:B10 (the Type 1 values)
- Type ,
- Specify the criteria (the cell E3, which has the value "Grass")
- Type ,
- Specify the range for the sum C2:C10 (the Total values)
- Type ,
- Hit enter

	A	B	C	D	E	F	G	H	I
1	Name	Type 1	Total						
2	Bulbasaur	Grass	318		Type	Total Sum			
3	Ivysaur	Grass	405		Grass	=SUMIF(B2:B10; E3; C2:C10)			
4	Venusaur	Grass	525		Fire	SUMIF (range; criteria; [sum_range])			
5	Charmander	Fire	309		Water				
6	Charmeleon	Fire	405						
7	Charizard	Fire	534						
8	Squirtle	Water	314						
9	Wartortle	Water	405						
10	Blastoise	Water	530						
11									

The function now sums the Total stats for Grass type Pokemon.

The function can be repeated for the other Pokemon types to compare them:

	A	B	C	D	E	F	G
1	Name	Type 1	Total				
2	Bulbasaur	Grass	318		Type	Total Sum	
3	Ivysaur	Grass	405		Grass	=SUMIF(B2:B10; E3; C2:C10)	
4	Venusaur	Grass	525		Fire	=SUMIF(B2:B10; E4; C2:C10)	
5	Charmander	Fire	309		Water	=SUMIF(B2:B10; E5; C2:C10)	
6	Charmeleon	Fire	405			SUMIF (range; criteria; [sum_range])	
7	Charizard	Fire	534				
8	Squirtle	Water	314				
9	Wartortle	Water	405				
10	Blastoise	Water	530				
11							

Now, we can see the sum of total stats for the different types of Pokemon:

The function can be repeated for the other Pokemon types to compare them:

	A	B	C	D	E	F	G
1	Name	Type 1	Total				
2	Bulbasaur	Grass	318		Type	Total Sum	
3	Ivysaur	Grass	405		Grass	=SUMIF(B2:B10; E3; C2:C10)	
4	Venusaur	Grass	525		Fire	=SUMIF(B2:B10; E4; C2:C10)	
5	Charmander	Fire	309		Water	=SUMIF(B2:B10; E5; C2:C10)	
6	Charmeleon	Fire	405			SUMIF (range; criteria; [sum_range])	
7	Charizard	Fire	534				
8	Squirtle	Water	314				
9	Wartortle	Water	405				
10	Blastoise	Water	530				
11							

	A	B	C	D	E	F	G
1	Name	Type 1	Total				
2	Bulbasaur	Grass	318		Type	Total Sum	
3	Ivysaur	Grass	405		Grass	1248	
4	Venusaur	Grass	525		Fire	1248	
5	Charmander	Fire	309		Water	1249	
6	Charmeleon	Fire	405				
7	Charizard	Fire	534				
8	Squirtle	Water	314				
9	Wartortle	Water	405				
10	Blastoise	Water	530				
11							

Now, we can see the sum of total stats for the different types of Pokemon:

	A	B	C	D	E	F	G
1	Name	Type 1	Total				
2	Bulbasaur	Grass	318		Type	Total Sum	
3	Ivysaur	Grass	405		Grass	1248	
4	Venusaur	Grass	525		Fire	1248	
5	Charmander	Fire	309		Water	1249	
6	Charmeleon	Fire	405				
7	Charizard	Fire	534				
8	Squirtle	Water	314				
9	Wartortle	Water	405				
10	Blastoise	Water	530				
11							

AVERAGE Function

The AVERAGE function is a premade function in Excel, which calculates the average (arithmetic mean).

It is typed =AVERAGE

It adds the range and divides it by the number of observations.

Example:

The average of (2, 3, 4) is 3.

3 observations (2, 3 and 4)

The sum of the observations ($2 + 3 + 4 = 9$)

($9 / 3 = 3$)

The average is 3

How to use the =AVERAGE function, step by step:

1. Select a cell (F2)
2. Type =AVERAGE
3. Double click the AVERAGE command
4. Select a range (B2:E2)
5. Hit enter
6. Next, Fill F2:F7

F2						
fx						
	A	B	C	D	E	F
1	Trainer	Pokeball	Great ball	Ultraball	Master ball	Average
2	Iva	10	4	1	1	
3	Liam	12	3	0	1	
4	Jenny	15	1	3	1	
5	Iben	4	2	6	0	
6	Adora	10	4	1	1	
7	Kasper	9	2	1	0	

MEDIAN Function

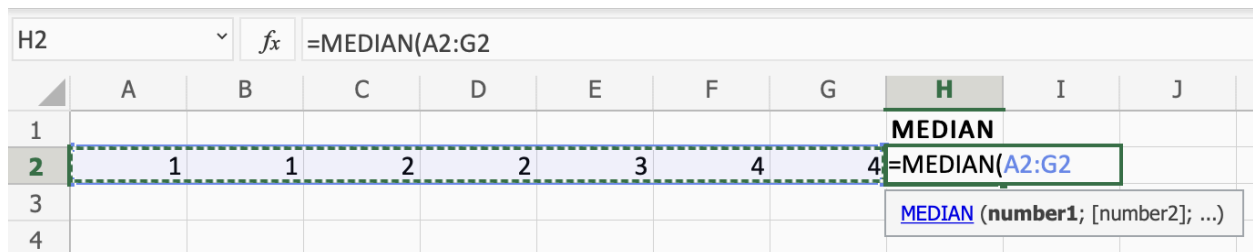
The MEDIAN function is a premade function in Excel, which returns the middle value in the data.

It is typed `=MEDIAN`

Calculating the median manually requires that the data is sorted and arranged from low to high. This is not necessary using the MEDIAN function, it does it for you.

How to use the `=MEDIAN` function, step by step:

1. Select a cell (H2)
2. Type `=MEDIAN`
3. Double click the MEDIAN command
4. Select a range (A2:G2)
5. Hit enter



	A	B	C	D	E	F	G	H	I	J
1								MEDIAN		
2	1	1	2	2	3	4	4	=MEDIAN(A2:G2)		
3										
4										

MODE Function

The MODE function is a premade function in Excel, which is used to find the number seen most times.

This function always returns a single number.

It is typed `=MODE.SNGL`

It returns the most occurring number in a range or array.

Let's have a look at an example where we help the Pokemon trainers to calculate the mode of numbers of Pokeballs.

How to use the `=MODE` function, step by step:

1. Select a cell (B10)
2. Type `=MODE`
3. Double click the MODE.SNGL command

4. Select a range (B2:E7)
5. Hit enter

B10		fx			
	A	B	C	D	E
1	Trainer	Pokeball	Great ball	Ultraball	Master ball
2	Iva	10	4	5	1
3	Liam	12	3	0	1
4	Jenny	15	1	3	1
5	Iben	4	2	2	0
6	Adora	1	4	2	1
7	Kasper	9	2	1	0
8					
9					
10	MODE				

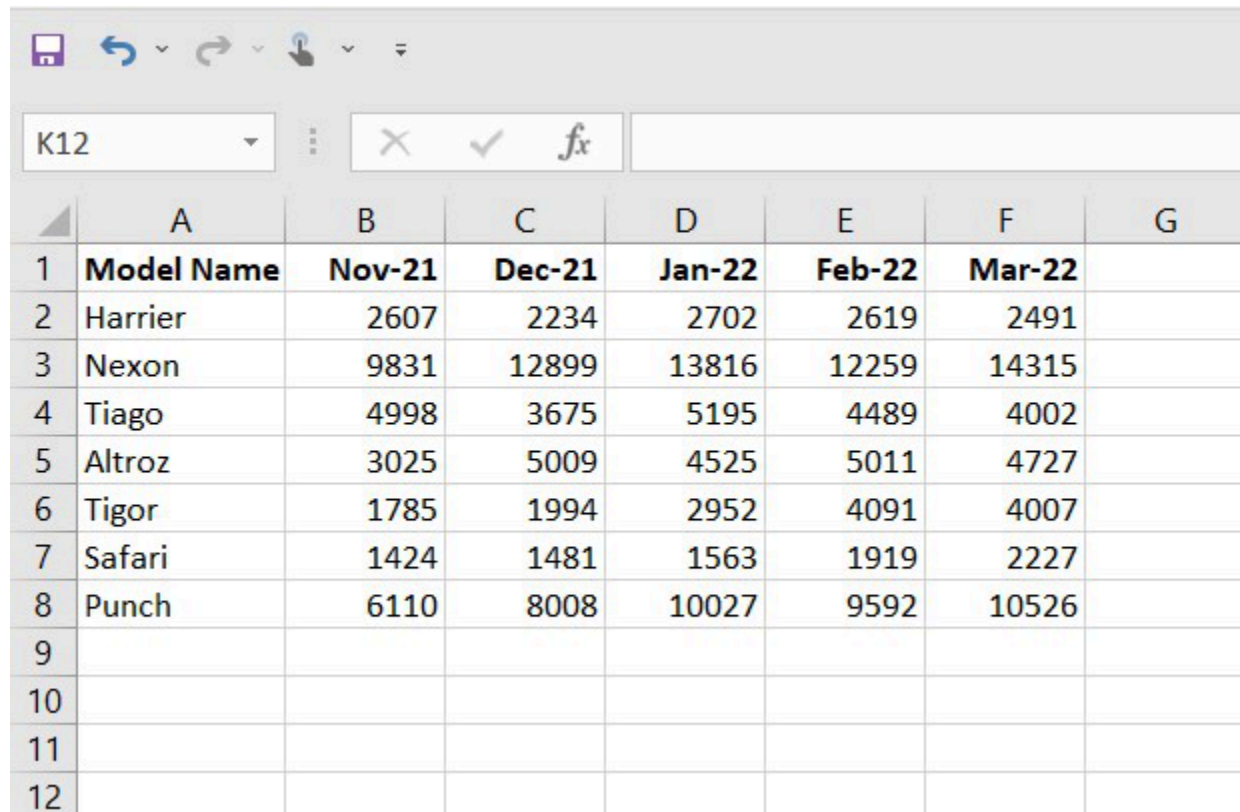
Types of Charts in Excel

Charts are used to represent the data into graphical elements, which makes it very to easy to interpret the data; it becomes very useful if our data is very large. Excel provides various charts to represent the excel data and makes it very easy to understand and analyze the data compare to the excel cells data analysis. There are various charts available in excel. Namely, they are,

- Line Chart
- Bar Chart
- Column Chart
- Area Chart
- Pie Chart

- Surface Chart

Create dataset

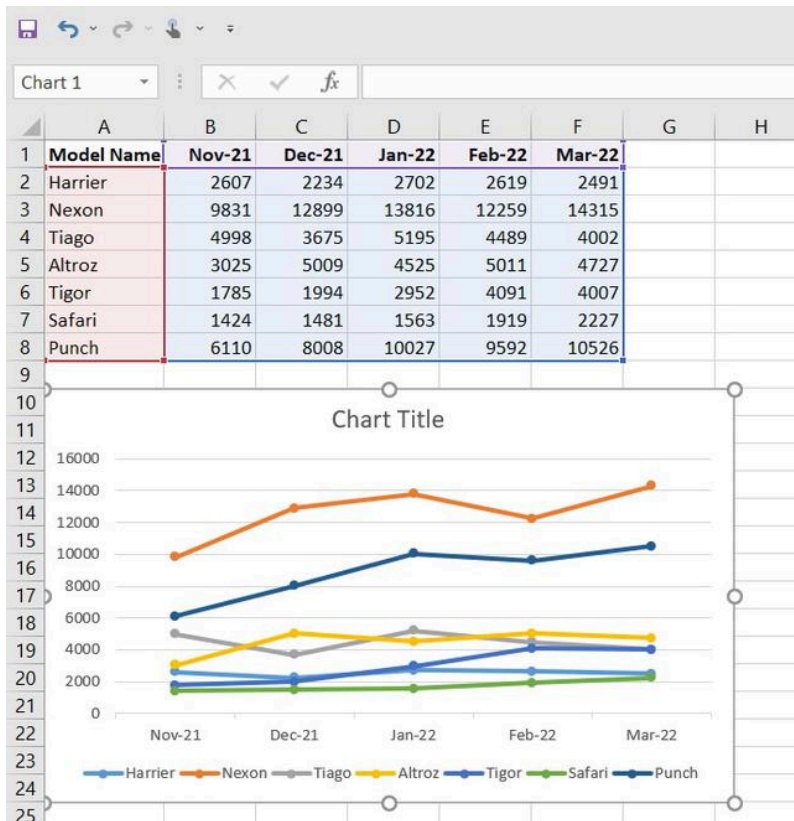


The screenshot shows a spreadsheet application with a toolbar at the top containing icons for saving, undo, redo, and a hand tool. Below the toolbar is a formula bar with a dropdown menu showing 'K12', a 'Cancel' button (X), a 'Confirm' button (checkmark), and a 'Function' button (fx). The main area of the spreadsheet is a table with 12 rows and 8 columns. The columns are labeled A through G. Row 1 contains headers: 'Model Name' in column A, and 'Nov-21', 'Dec-21', 'Jan-22', 'Feb-22', and 'Mar-22' in columns B through F respectively. Rows 2 through 8 contain data for car models: Harrier, Nexon, Tiago, Altroz, Tigor, Safari, and Punch. Rows 9 through 12 are empty.

	A	B	C	D	E	F	G
1	Model Name	Nov-21	Dec-21	Jan-22	Feb-22	Mar-22	
2	Harrier	2607	2234	2702	2619	2491	
3	Nexon	9831	12899	13816	12259	14315	
4	Tiago	4998	3675	5195	4489	4002	
5	Altroz	3025	5009	4525	5011	4727	
6	Tigor	1785	1994	2952	4091	4007	
7	Safari	1424	1481	1563	1919	2227	
8	Punch	6110	8008	10027	9592	10526	
9							
10							
11							
12							

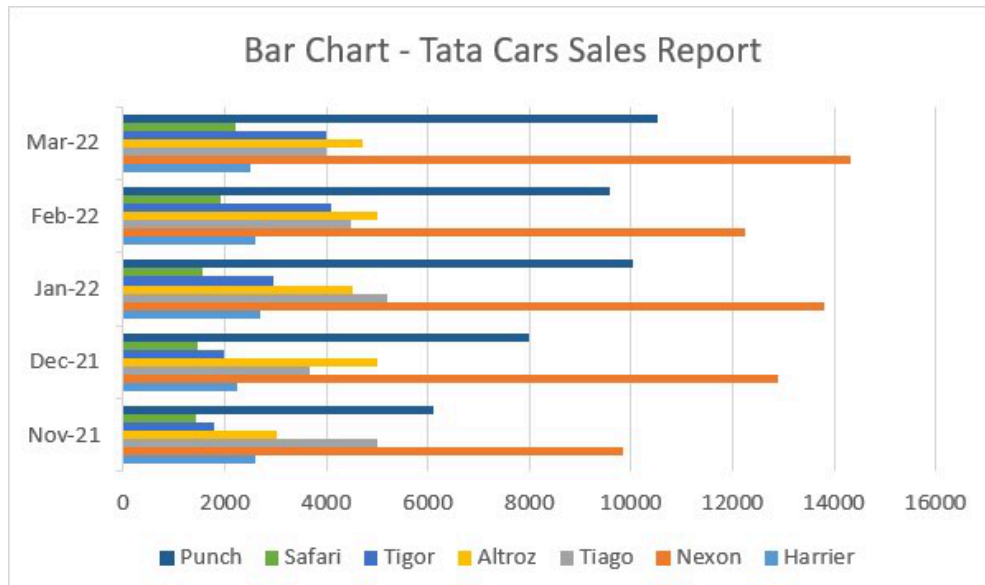
Line Chart

Line charts are most helpful in representing the trends. This can be very useful to analyze the ups and downs in a range of data over a particular time span. The data points in the chart are connected with the lines.



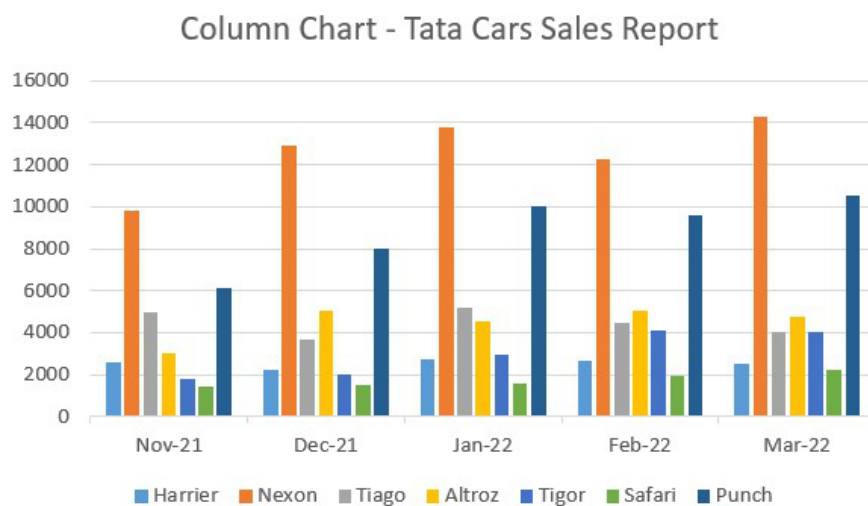
Bar Chart

Bar charts are used to represent the categorical data using the rectangular horizontal bars with their height and length proportional to the data values it is used to represent.



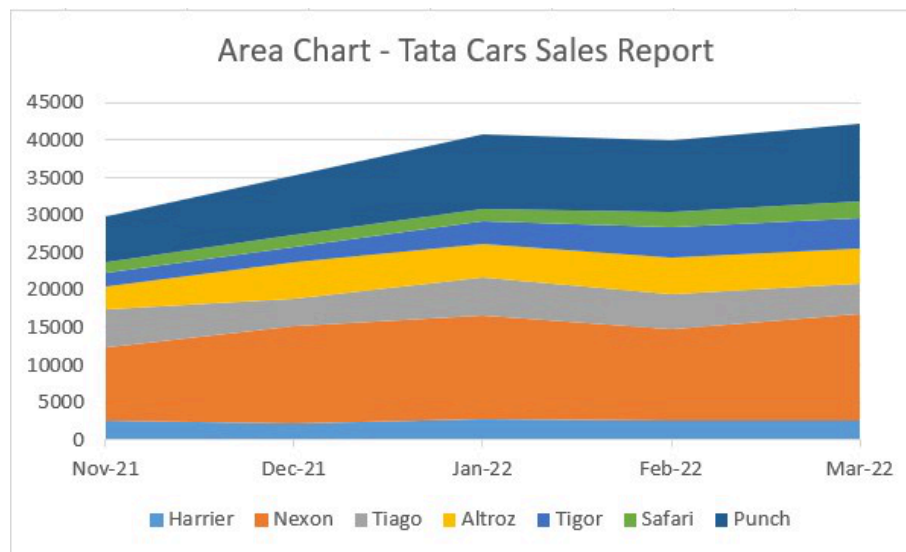
Column Chart

Column charts are used to represent the data in a vertical chart using the vertical bars. These graphs are mostly used for comparing the data points in the data. Column chart



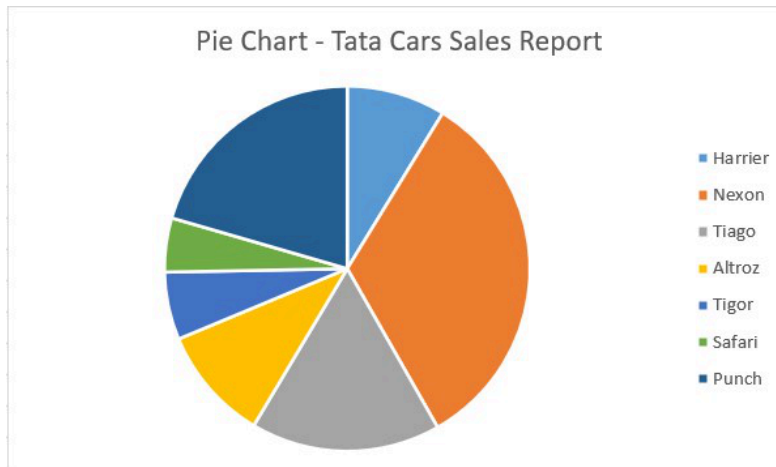
Area Chart

Area charts are used to display graphically quantitative data. It is similar to the line chart and based on it. The area between the lines is filled with color, and they are easy to analyze as they are similar to the line chart showing ups and downs in the data.



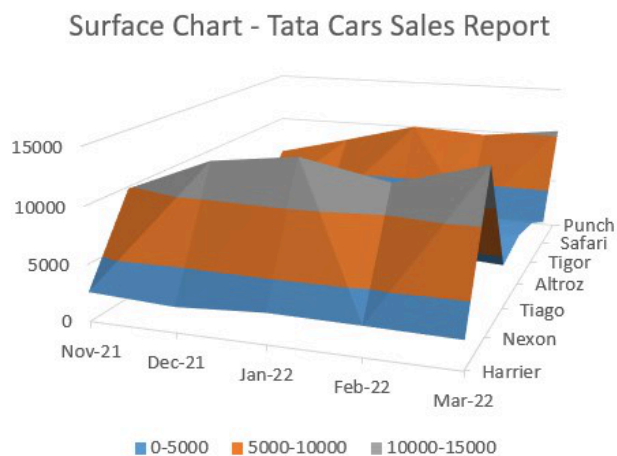
Pie Chart

Pie charts are circular statistical graphs that are divided into slices of pie in the proportion to data values to represent the data. They are commonly used to analyze the percentage allocation of data points in a complete dataset.



Surface Chart

Surface charts are 3-dimensional charts that are used to represent the data in a 3-dimensional landscape. They are mainly used to represent the large dataset. They display a variety of data at the same time.



How to use Text Functions in Excel

Excel is mostly about the numerical data, but at times you can come across the data which has too much text and that is the time when Text Functions in Excel will help you to simplify the things easily.

1.Left()

You can use the Left function when you want to extract the leftmost characters from a string. Syntax =left(text, num_char)

	B1		<i>f_x</i>
			=LEFT(A1,5)
	A	B	C
1	Learn Excel Easily	Learn	
2			
3			
4			

Similarly, you can also use the Right function to extract the rightmost characters from a string.

B1			=RIGHT(A1,5)	
	A	B	C	D
1	Learn Excel Easily	asily		
2				
3				
4				
5				

2. Len ()

Len function in Excel helps you to know the length of a string that is number of characters in a string. Syntax = LEN(text)

B1		f_x	=LEN(A1)	
	A	B	C	D
1	Learn Excel Easily	18		
2				
3				
4				
5				

3. Mid ()

Mid function in Excel is used to extract the characters from the middle of a string. Syntax = MID(text, start_char, num_chars)

B1					f_x	=MID(A1,7,5)
	A	B	C	D		
1	Learn Excel Easily	Excel				
2						
3						
4						
5						

4. Find ()

Find function in Excel is used when you want to know the position of certain characters in a particular string. Syntax =FIND(find_text, within_text,[start_num])

B1		fx =FIND("Easi",A1)			
	A	B	C	D	
1	Learn Excel Easily	13			
2					
3					
4					
5					

5. Proper ()

Proper function in Excel capitalizes each word in the string that is, it converts the case into proper case. Syntax =PROPER(Text)

B1		fx =PROPER(A1)			
	A	B	C	D	
1	learn excel easily	Learn Excel Easily			
2					
3					
4					
5					

6. Rept ()

Rept function in Excel is used when you want a certain text to be repeated certain number of times. Syntax =REPT(Text, number_times)

B1		fx =REPT(A1,2)		
	A	B	C	
1	learn excel easily	learn excel easylearn excel easily		
2				
3				
4				
5				
6				

7. Trim()

Trim function in Excel removes the unnecessary spaces from a particular string. Syntax =TRIM(Text)

B1		f_x =TRIM(A1)
	A	B
1	learn excel easily	learn excel easily
2		
3		
4		
5		

8. Upper()

Upper function in Excel converts the text into Upper case from lower case. Syntax =UPPER(Text)

B1		f_x =UPPER(A1)
	A	B
1	learn excel easily	LEARN EXCEL EASILY
2		
3		
4		
5		

9. Substitute ()

Substitute function in Excel helps to replace existing text with a new text in a particular string. Syntax =SUBSTITUTE(text, old_text, new_text, instance number)

B1		f_x =SUBSTITUTE(A1,"learn","Understand")
	A	B
1	learn excel easily	Understand excel easily
2		
3		
4		
5		

10. Concatenate ()

Concatenate function in Excel helps to join the text of two or more cells. Syntax =CONCATENATE(text1, text2&€¦.)

D1		fx		=CONCATENATE(A1,B1,C1)	
	A	B	C	D	E
1	Learn	Excel	Easily	LearnExcelEasily	
2					
3					
4					
5					

DateTime Function

The DATE function is used to calculate dates in Excel. To work with these formulas you first need to insert the dates in Excel. Excel provides different functions to work with dates & times such as TODAY, NOW, WEEKDAY, EOMONTH, etc. which we will discuss here with examples.

Excel DATE function List

DATE Function

Get the Current date and Time

TODAY- It returns today's date.

NOW- It returns the current date and time.

Converts Dates to / From text

DATEVALUE- Helps to Convert date in the text format to date format

TEXT- It helps to convert a date to a text value.

Retrieve dates in Excel

DAY- It returns the day of the month

MONTH- It is used to return the month of a specified date

YEAR- It is used to return the year of a specified date

EOMONTH- It returns the last day of the month.

WEEKDAY- It is used to return the day of the week.

WEEKNUM- It is used to return the week number of a date.

Q Write a formula to display today's date.

Ans: TODAY()

Q Write a formula to display the current date and time.

Ans: Now()

Q Extract year, Month and Day from 17/12/2024

Ans: Extract the year : =Year()

Extract the month: =month()

Extract the day : =day()

Q Write formula for create date

Ans : =date(year,month,date)

Q write formula for create time

Ans : =time(hour,min,sec)

Excel DATEVALUE Function

It converts the date in text format to a serial number, which can be represented as a date.

Ans" DATEVALUE(date_text)

B1		fx		=DATEVALUE("27/03/2021")	
	A		B		C
1			44282		

Excel TEXT Function

It converts any numeric value not only dates to a text string. Through this function, we can change the date to text strings in a variety of formats.

TEXT(value,format_text)

Example 1

B1		fx		=TEXT(A1,"d-mmmm-yyyy")	
	A		B		C
1	28-03-2021		28-March-2021		

Example 2

B1		fx		=TEXT(A1,"dd mmm,yyy")	
	A		B		C
1	28-03-2021		28 Mar,2021		

Example 3

B1		f_x	=TEXT(A1,"dddd,mmmm d ,yyy")
	A	B	C
1	31-03-2021	Wednesday, March 31 ,2021	

Excel DAY Function

It returns the day of a month, i.e. integer from 1 to 31.

Syntax:

DAY(serial_number)

B1		f_x	=DAY(A1)
	A	B	C
1	29-03-2021	29	

B1		f_x	=DAY(DATE(2021,3,31))
	A	B	
1		31	

The DAY(TODAY()) function returns the day of today's date, as shown below:

B1		f_x	=DAY(TODAY())
	A	B	C
1		1	

Excel MONTH Function

This function returns the month of the given date as an integer from 1 to 12 (January to December).

Syntax:

MONTH(serial_number)

serial_number: This value represents the date for which you want to find the month.

B1		f_x	=MONTH(A1)
	A	B	C
1	31-03-2021	3	

The MONTH(TODAY()) function returns the month of today's date.

B1		fx	=MONTH(TODAY())
	A	B	C
1			4

Excel WEEKDAY Function

This function returns the day of the week as a number from 1 to 7 (Sunday to Saturday) according to the specified date.

Syntax:

WEEKDAY(serial_number,return_type)

Arguments:

1. serial_number: It can be a date or the cell that contains the date.
2. return_type: It is optional, as it specifies which day should be considered as the first day of the week.

B2		fx	=WEEKDAY(A2)
	A	B	
1		RESULT	
2	30-03-2021		3
3			

Logical Functions in Excel

"Logical functions are used to compare more than one condition or multiple conditions. It returns the result as TRUE or FALSE by evaluating the arguments."

These functions are used for calculating the result and help to elect any one of the given data. Based on the requirement, the contents in the cell are evaluated using the respective logical condition. Here in this tutorial, the types of Logical Functions used are,

- AND
- OR
- NOT

- XOR

as it is used to evaluate the result.

AND Function

The AND function tests single or multiple conditions. It returns the value true if all the values evaluate to true and return false if any one of the value evaluates to false. This condition evaluates more than one condition, accepting 255 conditions called arguments. Arguments include expressions, constants, arrays, and cell references.

Syntax

Example 1: Evaluate the given data using AND condition

To evaluate the data using certain conditions, the steps to be followed are

1. Enter the data in the worksheet namely A1:A5
2. Here, the data present in A1:A5 are calculated as greater than 1 and lesser than 100.
3. Select a new cell and enter the formula as = AND (A1:A5>1, A1:A5<100). Press Enter.
4. The result will be displayed as either TRUE or FALSE in the selected cell.

B1		fx =AND(A1:A5>1,A1:A5<100)					
	A	B	C	D	E	F	G
1	10	TRUE					
2	40	TRUE					
3	50	TRUE					
4	60	TRUE					
5	70	TRUE					

The result in the worksheet will be displayed as TRUE as the condition evaluates to true.

IF and AND Function

IF and AND Function

The IF and AND functions are used together for multiple conditions.

Example 2: Evaluate the given data using IF and AND function

- | | | | | | | | | | | |
|----|----|---|---|---|---|---|---|---|---|--|
| B1 | | =IF(AND(A1<A2,A1<100),A1,"The value is out of range") | | | | | | | | |
| | A | B | C | D | E | F | G | H | I | |
| 1 | 10 | 10 | | | | | | | | |
| 2 | 40 | | | | | | | | | |
| 3 | | | | | | | | | | |

The two cell values are compared using IF and AND functions.

1. Enter the data in the worksheet, namely A1
2. Here, the data in A1 and the value in A1 are displayed if $A1 > 1$ and $A1 < 100$. If the condition is false, it displays the message "The value is out of range".
3. Select a new cell and enter the formula as = IF (AND (A1>1, A1<100), A1, "The value is out of range"). Press Enter.
4. The result will be displayed as either the value present in cell A1 or "The value is out of range"

Figure 10-10: The IF function with AND

OR Function

The OR function returns the result as True if any arguments evaluate to true and return False if all the arguments evaluate to False. It acts on multiple testing

conditions. It is combined with AND function and IF condition based on the requirement.

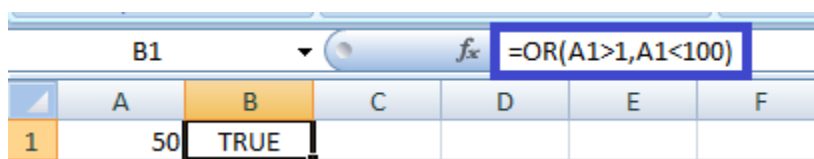
Syntax

=OR (logical 1, [logical 2],...)

Example 1: Evaluate the given data using OR function

To evaluate the data using OR condition, the steps to be followed are:

1. Enter the data in the worksheet, namely A1.
2. Here, the data present in A1 are calculated, which is greater than 1 and lesser than 100.
3. Select a new cell and enter the formula as = OR (A1>1, A1<100). Press Enter.
4. The result will be displayed as either TRUE or FALSE in the selected cell.



	A	B	C	D	E	F
1	50	TRUE				

In the worksheet, the result will be displayed as TRUE where the arguments present in the condition evaluate to true. In example 1, the arguments are based on a single cell.

NOT Function

NOT is one of the logical functions which return the reversed logical value. An inbuilt function in Excel is used along with the formula based on the requirement.

Syntax

=NOT (logical value)

Logical value- An expression is used to evaluate either TRUE or FALSE

As described before, the NOT function reverses the value of the argument. Therefore, if the logical value is TRUE, the result will be false, and if the logical value is FALSE, the result will be TRUE.

Lookups :

Lookup functions in Excel are used for looking through a single column or row to find a particular value from the same place in a second column or row. This often takes place when there are multiple worksheets within a workbook or a large amount of data in a worksheet. Lookup functions are very helpful with large sets of data to find specific information across many rows or columns.

VLOOKUP (Vertical Lookup)

Looks for a value in the leftmost column of a table and returns a value in the same row from the vertical column you define. The lookup value must always be in the leftmost column of the table.

The Excel formula is written

=VLOOKUP(lookup_value,table_array,col_index_num,[range_lookup])

This table includes the sales totals for salespeople. In this case you want to find out what the sales were for certain salespeople. This is a small example but imagine a long list of names and numbers to search through. Using the VLOOKUP function quickly brings you the results you want. Be sure to use F4 to create the absolute references and keep the table array the same for each lookup. Also make sure and make the col_index_num use the second column in the table (column 2 in the table) with a return for the range_lookup as FALSE. By using FALSE in all cases, there is no need to sort your tables into ascending order. Here is what those results look like.

IF		X		✓		fx		=VLOOKUP(A5,\$E\$5:\$F\$16,2,FALSE)	
	A	B	C	D	E	F	G	H	
1	VLOOKUP								
2									
3									
4	Value	Returns		Name	Sales				
5	Cortana	=VLOOKUP(A5,\$E\$5:\$F\$16,2,FALSE)							
6	Drago	VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])							
7	Sara			Cortana	798				
8	Evi			Robin	541				
9				Drago	483				
10				Smarti	661				
11				Vokul	178				
12				Tolt	275				
13				Evi	888				
14				Jeannie	321				
15				Sara	654				
16				Donna	987				
17									

C5							
	A	B	C	D	E	F	G
1	VLOOKUP						
2							
3							
4	Value		Returns		Name	Sales	
5	Cortana		798		Siri	933	
6	Drago		483		Alexa	362	
7	Sara		654		Cortana	798	
8	Evi		888		Robin	541	
9					Drago	483	
10					Smarti	661	
11					Vokul	178	
12					Tolt	275	
13					Evi	888	
14					Jeannie	321	
15					Sara	654	
16					Donna	987	

HLOOKUP (Horizontal Lookup)

Looks for a value in the topmost row of a table and returns a value in the same row from the horizontal column you define. The lookup value must always be in the leftmost column of the table.

The Excel formula is written

=HLOOKUP(lookup_value,table_array,row_index_num,[range_lookup])